

MTE 203 – Advanced Calculus

Week 4 Homework - Solutions

Tangential and Normal Components of Acceleration (S.11.13)

Problem 1: [11.13, Prob. 25]

Calculate the normal component a_N of the acceleration of a particle using equations 11.112 and 11.113 if its position is given by $x = t^2 + 1, y = 2t^2 - 1, z = t^2 + 5t, t \geq 0$, (t being time).

Note:

Equation 11.112 (Trim Book):

$$a_T = a \cdot \hat{T} = \frac{d}{dt}|v|, a_N = a \cdot \hat{N} = |v| \left| \frac{d\hat{T}}{dt} \right|$$

Equation 11.113 (Trim Book):

$$a_N = \sqrt{|a|^2 - a_T^2}$$

Solution:

$$\text{Since } \hat{T} = \frac{(2t, 4t, 2t + 5)}{\sqrt{4t^2 + 16t^2 + (2t + 5)^2}} = \frac{(2t, 4t, 2t + 5)}{\sqrt{24t^2 + 20t + 25}},$$

$$\begin{aligned} \frac{d\hat{T}}{dt} &= \frac{-(24t + 10)}{(24t^2 + 20t + 25)^{3/2}}(2t, 4t, 2t + 5) + \frac{(2, 4, 2)}{\sqrt{24t^2 + 20t + 25}} \\ &= \frac{1}{(24t^2 + 20t + 25)^{3/2}} [-(24t + 10)(2t, 4t, 2t + 5) + (24t^2 + 20t + 25)(2, 4, 2)] \\ &= \frac{10(2t + 5, 4t + 10, -10t)}{(24t^2 + 20t + 25)^{3/2}}. \end{aligned}$$

According to 11.112b,

$$\begin{aligned} a_N &= |(2t, 4t, 2t + 5)| \left| \frac{10(2t + 5, 4t + 10, -10t)}{(24t^2 + 20t + 25)^{3/2}} \right| \\ &= \sqrt{24t^2 + 20t + 25} \left[\frac{10\sqrt{(2t + 5)^2 + (4t + 10)^2 + 100t^2}}{(24t^2 + 20t + 25)^{3/2}} \right] \end{aligned}$$

$$= \frac{10\sqrt{5}}{\sqrt{24t^2 + 20t + 25}}.$$

Since $\mathbf{v} = (2t, 4t, 2t + 5)$ and $\mathbf{a} = (2, 4, 2)$,

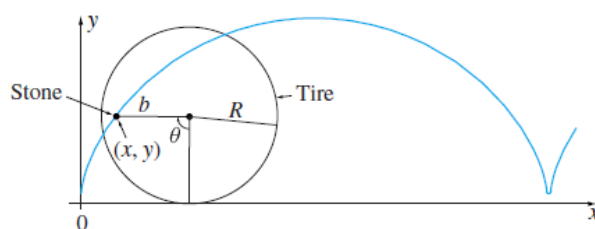
$$a_T = \frac{d}{dt}|\mathbf{v}| = \frac{d}{dt}\sqrt{24t^2 + 20t + 25} = \frac{24t + 10}{\sqrt{24t^2 + 20t + 25}}.$$

Equation 11.113 gives $a_N = \sqrt{24 - \frac{(24t + 10)^2}{24t^2 + 20t + 25}} = \frac{10\sqrt{5}}{\sqrt{24t^2 + 20t + 25}}.$

Problem 2: [11.13, Prob. 39]

If a stone is embedded in the side of the tire, its path is called a trochoid (see Prob. 58 in section 9.1). If the distance from the center of the tire to the stone is b , parametric equations for the trochoid are,

$$x = R\theta - b \sin \theta, \quad y = R - b \cos \theta$$



- Find the velocity, speed, and acceleration of the stone if the tire rolls so that its center has constant speed S . (Assume the $x = 0$ at time $t = 0$.)

Solution:

Since the length around the circumference of the tire is given by $R\theta$ and the rate of change of this quantity with respect to time is the speed of the center of the tire S , it follows that

$$S = R \left(\frac{d\theta}{dt} \right)$$

Integrating this expression,

$$\int S dt = \int R d\theta$$

$$\theta = \frac{S}{R}t + C$$

Where, C is a constant. Since $\theta = 0$ when $t = 0$, it follows that $C = 0$, and

$$\theta = \frac{S}{R}t$$

From the parametric equations of the trochoid the vector function for this motion can be written as,

$$\vec{r}(t) = (R\theta - b \sin \theta)\hat{i} + (R - b \cos \theta)\hat{j}$$

Using the chain rule, we can then calculate the velocity as,

$$\begin{aligned}\mathbf{v} &= \frac{d\mathbf{r}}{dt} = \left(R\frac{d\theta}{dt} - b \cos \theta \frac{d\theta}{dt}\right)\hat{i} + \left(b \sin \theta \frac{d\theta}{dt}\right)\hat{j} = \frac{S}{R} \left[(R - b \cos \theta)\hat{i} + b \sin \theta \hat{j}\right] \\ |\mathbf{v}| &= \frac{S}{R} \sqrt{(R - b \cos \theta)^2 + b^2 \sin^2 \theta} = \frac{S}{R} \sqrt{R^2 - 2Rb \cos \theta + b^2} \\ \mathbf{a} &= \frac{d\mathbf{v}}{dt} = \frac{S}{R} \left[\left(b \sin \theta \frac{d\theta}{dt}\right)\hat{i} + \left(b \cos \theta \frac{d\theta}{dt}\right)\hat{j}\right] = \frac{bS^2}{R^2} (\sin \theta \hat{i} + \cos \theta \hat{j}).\end{aligned}$$

b. What are the normal and tangential components of the stone's acceleration?

Solution:

Once again, using the chain rule

$$\begin{aligned}a_T &= \frac{d}{dt}|\mathbf{v}| = \frac{S}{R} \frac{2Rb \sin \theta (d\theta/dt)}{2\sqrt{R^2 - 2Rb \cos \theta + b^2}} = \frac{bS^2}{R} \frac{\sin \theta}{\sqrt{R^2 - 2Rb \cos \theta + b^2}} \\ a_N &= \sqrt{|\mathbf{a}|^2 - a_T^2} = \left(\frac{b^2 S^4}{R^4} - \frac{b^2 S^4}{R^2} \frac{\sin^2 \theta}{R^2 - 2Rb \cos \theta + b^2}\right)^{1/2} = \frac{bS^2}{R^2} \left(1 - \frac{R^2 \sin^2 \theta}{R^2 - 2Rb \cos \theta + b^2}\right)^{1/2} \\ &= \frac{bS^2}{R^2} \left[\frac{R^2 - 2Rb \cos \theta + b^2 - R^2(1 - \cos^2 \theta)}{R^2 - 2Rb \cos \theta + b^2}\right]^{1/2} \\ &= \frac{bS^2}{R^2} \left[\frac{(-b + R \cos \theta)^2}{R^2 - 2Rb \cos \theta + b^2}\right]^{1/2} = \frac{bS^2}{R^2} \frac{|-b + R \cos \theta|}{\sqrt{R^2 - 2Rb \cos \theta + b^2}}\end{aligned}$$

Multivariable Functions (S.12.1)

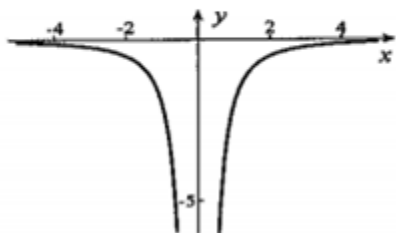
Problem 1: [12.1, Prob. 5]

Find and illustrate geometrically the largest possible domain for the function:

$$f(x, y) = \sin^{-1}(x^2 y + 1)$$

Solution:

For $-1 \leq x^2y + 1 \leq 1$, we require
 $-2 \leq x^2y \leq 0$ or $-2/x^2 \leq y \leq 0$. Points are
below the x -axis and above the curve $y = -2/x^2$.
Points on the boundary are also included.

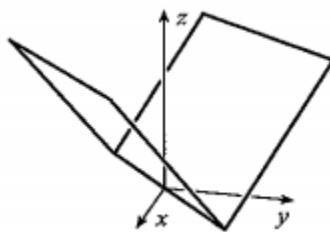


Problem 2: [12.1, Prob. 19]

Draw the surface defined by the following function:

$$f(x, y) = |x - y|$$

Solution:

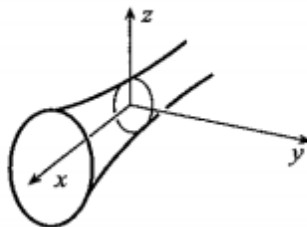


Problem 3: [12.1, Prob. 21]

Draw the surface defined by the following function:

$$f(x, y) = \sqrt{1 + x^2 - y^2}$$

Solution:



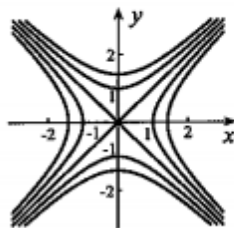
Problem 4: [12.1, Prob. 25]

Draw the level curves $f(x, y) = C$ corresponding to the values $C = -2, -1, 0, 1, 2$ for the curve below:

$$f(x, y) = x^2 - y^2$$

Solution:

Level curves are defined by $x^2 - y^2 = C$.
They are hyperbolas, except when $C = 0$
when they are the lines $y = \pm x$.



Problem 5: [S.12.1, Prob. 31]

A long piece of metal 1 m wide is bent in two places A and B (figure below) to form a channel with three straight sides. Find a formula (i.e. a multivariable function) for the cross-sectional area of the channel in terms of x , θ , and ϕ .

Note:

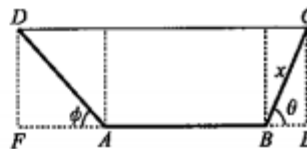
Defining a multivariable function for a physical problem like this one will be useful when we start solving optimization problems with constraints (e.g., maximizing the dimensions of a box to fit within a specific space)

Solution:

Since

$$\begin{aligned}\|DF\| &= \|CE\| = x \sin \theta, \\ \|AF\| &= \|DF\| \cot \phi = x \sin \theta \cot \phi, \\ \|AB\| &= 1 - \|BC\| - \|AD\| \\ &= 1 - x - x \sin \theta \csc \phi,\end{aligned}$$

the cross-sectional area is



$$\begin{aligned}\text{Area} &= \|AB\| \|CE\| + \frac{1}{2} \|BE\| \|CE\| + \frac{1}{2} \|FA\| \|DF\| \\ &= \|CE\| \left(\|AB\| + \frac{1}{2} \|BE\| + \frac{1}{2} \|FA\| \right)\end{aligned}$$

$$\text{Area}(x, \theta, \phi) = x \sin \theta \left[(1 - x - x \sin \theta \csc \phi) + \frac{1}{2} (x \cos \theta) + \frac{1}{2} (x \sin \theta \cot \phi) \right]$$

Partial Derivatives (S.12.3)

Problem 1: [12.3, Prob. 39]

The equation of continuity for three-dimensional unsteady flow of a compressible fluid is,

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x}(\rho u) + \frac{\partial}{\partial y}(\rho v) + \frac{\partial}{\partial z}(\rho w) = 0$$

Where $\rho(x, y, z, t)$ is the density of the fluid, and $u\hat{i} + v\hat{j} + w\hat{k}$, is the velocity of the fluid at position (x, y, z) and time t . Determine whether the continuity equation is satisfied if:

- a. $\rho = \text{constant}$, $u = (2x^2 - xy + z^2)t$, $v = (x^2 - 4xy + y^2)t$, $w = (-2xy - yz + y^2)t$
- b. $\rho = xy + zt$, $u = x^2y + t$, $v = y^2z - 2t^2$, $w = 5x + 2z$

Solution:

$$\begin{aligned} \text{(a)} \quad & \frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x}(\rho u) + \frac{\partial}{\partial y}(\rho v) + \frac{\partial}{\partial z}(\rho w) = 0 + \rho(4xt - yt) + \rho(-4xt + 2yt) + \rho(-yt) = 0 \\ \text{(b)} \quad & \frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x}(\rho u) + \frac{\partial}{\partial y}(\rho v) + \frac{\partial}{\partial z}(\rho w) = z + \frac{\partial}{\partial x}(x^3y^2 + xyt + x^2yzt + zt^2) \\ & + \frac{\partial}{\partial y}(xy^3z - 2xyt^2 + y^2z^2t - 2zt^3) + \frac{\partial}{\partial z}(5x^2y + 2xyz + 5xzt + 2z^2t) \\ & = z + (3x^2y^2 + yt + 2xyzt) + (3xy^2z - 2xt^2 + 2yz^2t) + (2xy + 5xt + 4zt) \neq 0 \end{aligned}$$

Problem 2: [12.3, Prob. 41b]

In complex variable theory, two functions $u(x, y)$ and $v(x, y)$ are said to be *harmonic conjugates* in a region R if in R they satisfy the Cauchy-Reimann equations

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$$

Show that the following pair of functions are harmonic conjugates:

$$u(x, y) = \frac{x^2 + x + y^2}{x^2 + y^2}, \quad v(x, y) = \frac{-y}{(x^2 + y^2)}$$

Solution:

$$\begin{aligned}\frac{\partial u}{\partial x} &= \frac{(x^2 + y^2)(2x + 1) - (x^2 + x + y^2)(2x)}{(x^2 + y^2)^2} = \frac{y^2 - x^2}{(x^2 + y^2)^2} \\ \frac{\partial u}{\partial y} &= \frac{(x^2 + y^2)(2y) - (x^2 + x + y^2)(2y)}{(x^2 + y^2)^2} = \frac{-2xy}{(x^2 + y^2)^2} \\ \frac{\partial v}{\partial x} &= \frac{2xy}{(x^2 + y^2)^2} \quad \frac{\partial v}{\partial y} = \frac{(x^2 + y^2)(-1) + y(2y)}{(x^2 + y^2)^2} = \frac{y^2 - x^2}{(x^2 + y^2)^2}\end{aligned}$$

Extra Practice Problems¹

The answers to these problems, along with all even-numbered exercises, are provided at the back of the textbook. Detailed solutions can be found in Trim's Student Solution Manual. If you don't own a copy, a reserved edition is available at the Davis Centre.

1. S. 11.13, Probs. 26, 38, 44
2. S. 12.1, Probs. 4, 18, 20, 22, 26
3. S. 12.3, Probs. 18, 24, 28, 32, 42

¹ You are strongly encouraged to try and solve the problems assigned in this homework on your own, before looking at the solutions. These problems are only the minimum necessary to master the material. You are also encouraged to do additional problems from the text for practice (some suggested **extra practice problems** have been listed here).